Department of Computer Science, Universita degli studi di Verona
Master’s Degree in Medical Bioinformatic

Comparative Evaluation of Bioinformatics Pipelines for Circular RNA Detection and Reconstruction

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Background – What are CircRNAs?

Definition & Key Properties:

•Circular RNAs (circRNAs) – covalently closed loops formed by back-splicing.

•Highly stable (resistant to exonucleases).

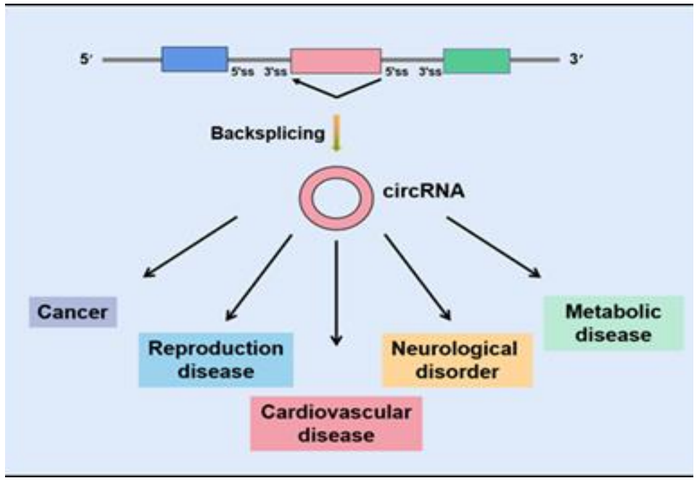
•Dysregulated in cancer, cardiovascular, and neurological diseases.

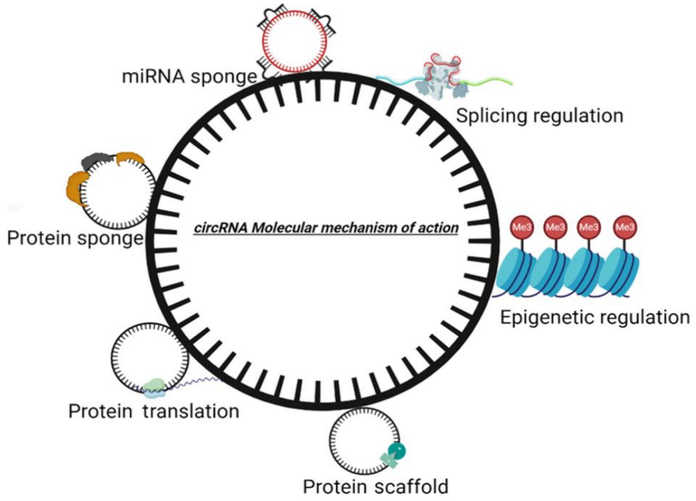
Biological Significance:

•miRNA sponges – bind and trap miRNAs, regulating gene expression.

•Protein scaffolds – bring together protein complexes for cellular signaling.

•Biomarker potential – stable in body fluids (blood, saliva) → ideal for liquid biopsies.





Motivation & Problem Statement

CURRENT LIMITATIONS

- Multiple detection tools exist, but each has its own algorithm, biases, and output format.
- Low agreement between different tools (benchmarking studies show **<20% overlap**).
- Researchers often run only one tool → risk of false positives or missed circRNAs.
- Manual execution of several tools is time-consuming, error-prone, and hard to reproduce.

THE SOLUTION

Why this matters:

"We need a way to combine multiple tools automatically and get a reliable set of high-confidence results."

✨ **AUTOMATED INTEGRATION**

Objective

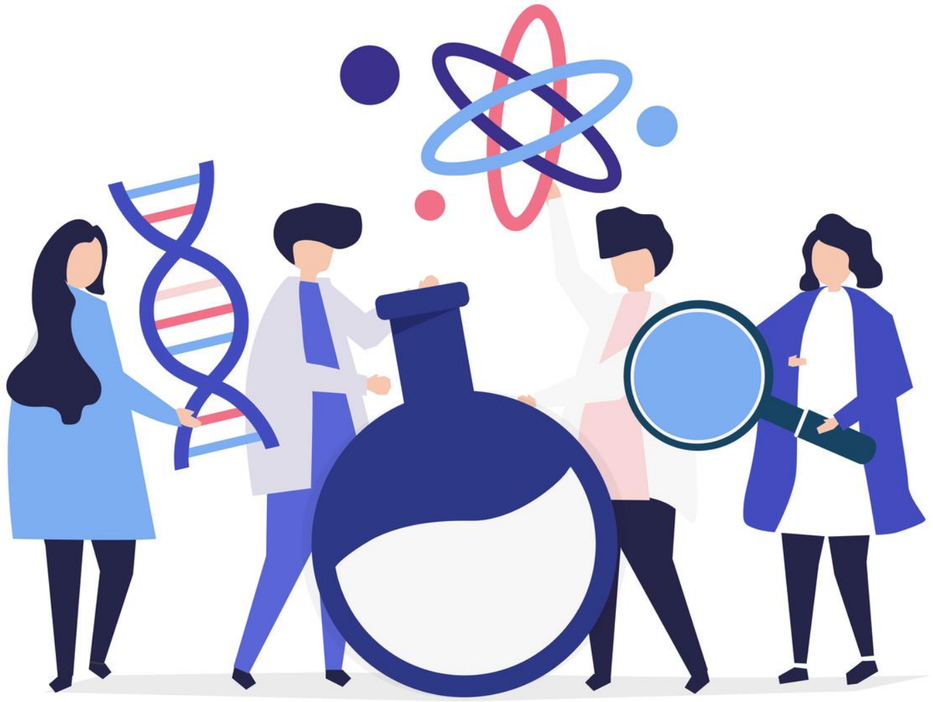
What we aimed to build

1.Automate the execution of four widely used circRNA detection pipelines
– CIRCexplorer2, CIRI-Full, CyCLeR, circtools

2.Standardize their outputs into a common format (coordinates, TPM)

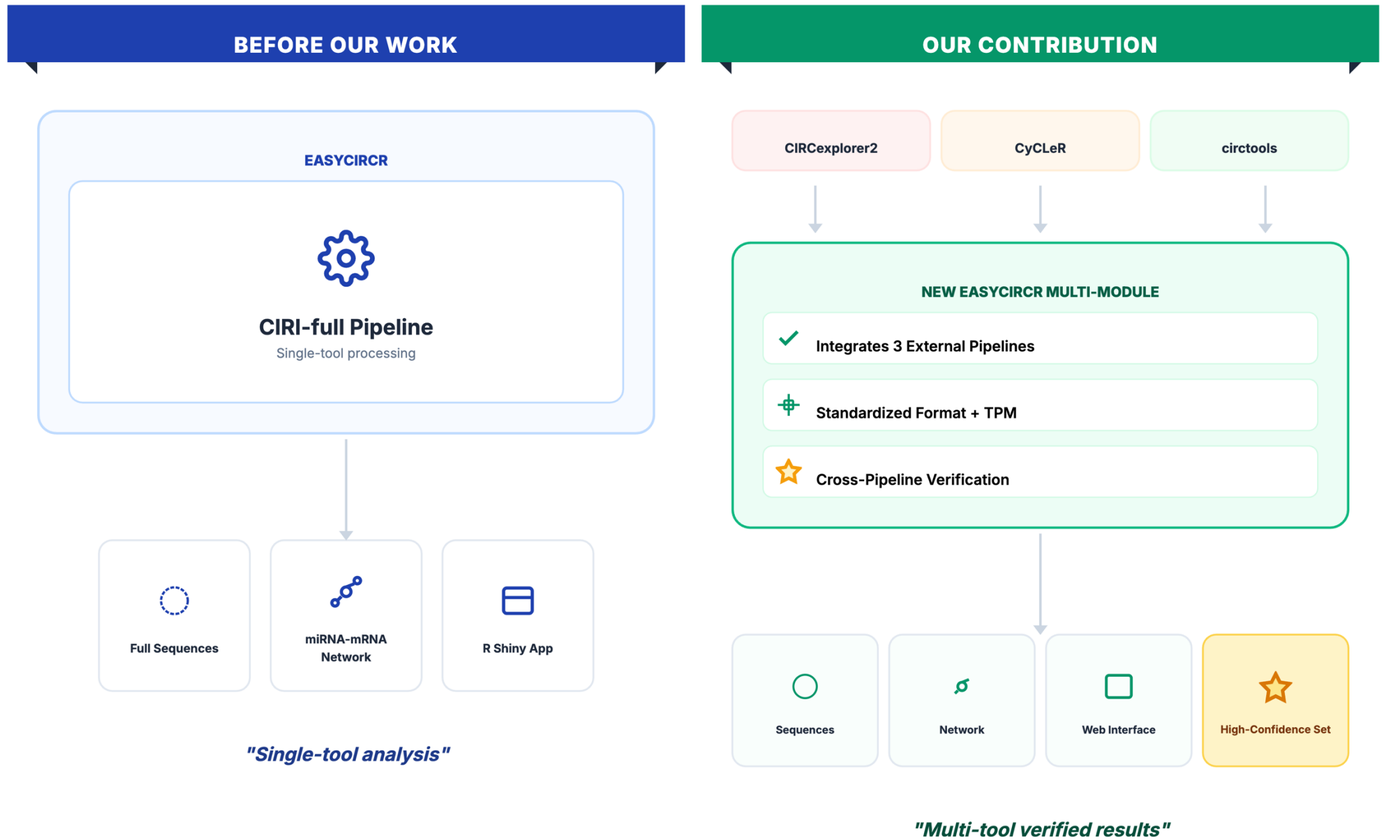
3.Integrate these standardized results into **EasyCircR**
– So users can compare tools, build consensus sets, and perform downstream analysis

4.Validate the approach by comparing detection and quantification across tools



What is EasyCircR?

- **EasyCircR** is an R package that makes circRNA analysis easier.



Methodology Overview



1. RNA-seq Data Acquisition

Collection of raw RNA sequencing data from relevant biological samples.



2. Quality Control & Preprocessing

Rigorous filtering of low-quality reads, adapter trimming, and data normalization.



3. Read Alignment

Mapping of clean reads to the reference genome (hg38) to identify genomic locations.



4. circRNA Detection

Identification of circRNA candidates using a diverse set of leading bioinformatics pipelines.



5. Comparative Analysis

Evaluation of detection accuracy, sensitivity, and overlap across all employed pipelines.



6. Validation & Recommendations

High-confidence circRNA identification and best practice pipeline recommendations.

Dataset & Experimental Setup

Diffuse Large B-Cell Lymphoma (DLBCL) Study Parameters



Data Source & Sample

TMD8-8h-DMSO: Cells treated with DMSO for 8 hours. High-throughput paired-end RNA-seq.



Reference Framework

- **Genome:** Human Reference Genome `hg38` (GRCh38)
- **Annotations:** UCSC GTF (Gene, Transcript, & Exon locations)

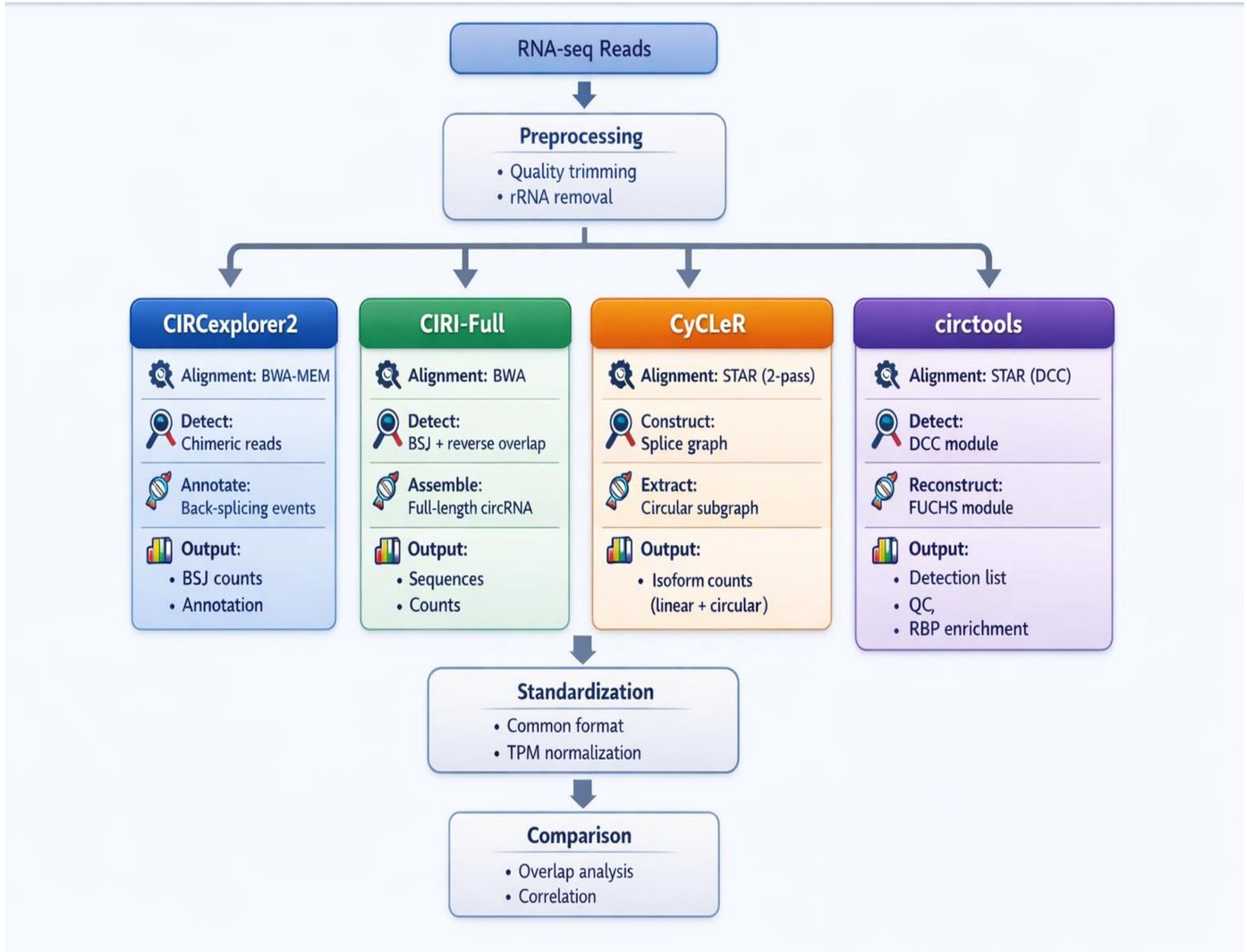


Computational Environment

High-performance computing (HPC) resources via University Server. Data analysis performed using R and Bash.

Material & Methods

- **CIRCexplorer2:** Finds circRNAs by detecting back-splice junctions from spliced reads
- **CIRI-Full:** Reconstructs full-length circRNA sequences using junction and reverse-overlap reads
- **CyCLeR:** Quantifies both circular and linear RNAs together using a graph-based approach
- **circtools:** A modular toolkit covering detection, QC, structure, and enrichment



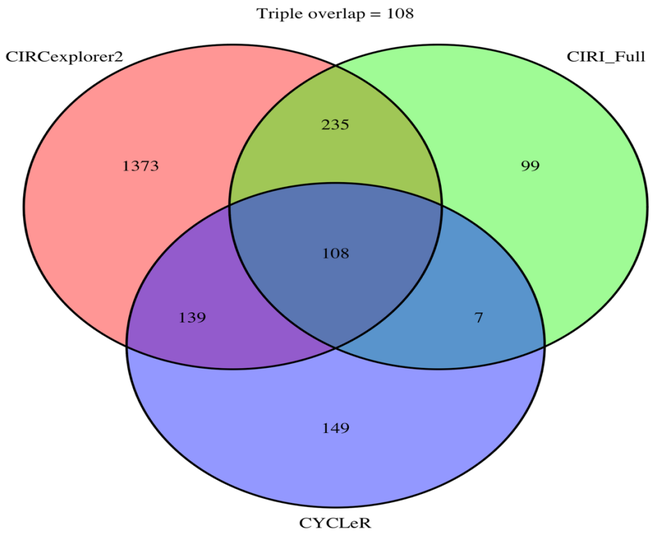
Material & Methods

Method Comparison

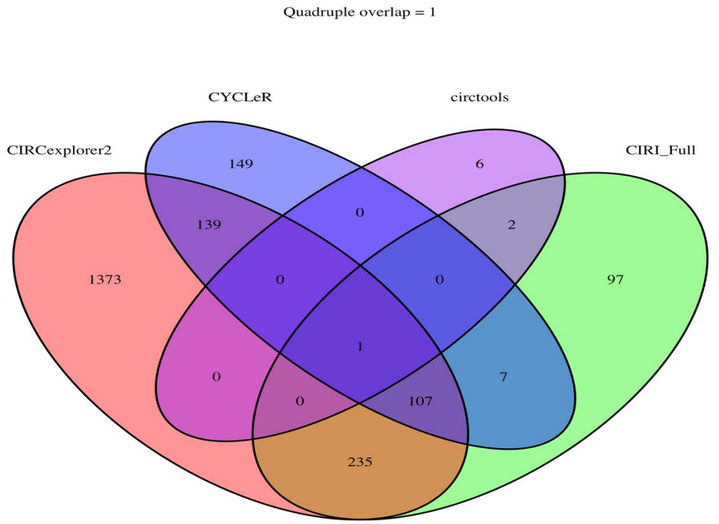
Tool	Strength	Limitation
CIRCexplorer2	High sensitivity, discovers many circRNAs	More false positives
CIRI-Full	Full-length reconstruction	Low sensitivity, misses some circRNAs
CyCLeR	Isoform-level accuracy, quantifies linear + circular	Complex model, requires two library types
circtools	High specificity, modular, functional analysis	Detects few circRNAs (very strict)

Results: Detection Overlap

- 108 circRNAs were detected by all three tools
- This set represents high-confidence candidates
- CIRCexplorer2 and CIRI-Full shared 235 circRNAs – moderate agreement
- CyCLeR showed limited overlap with both – reflects its different graph-based approach

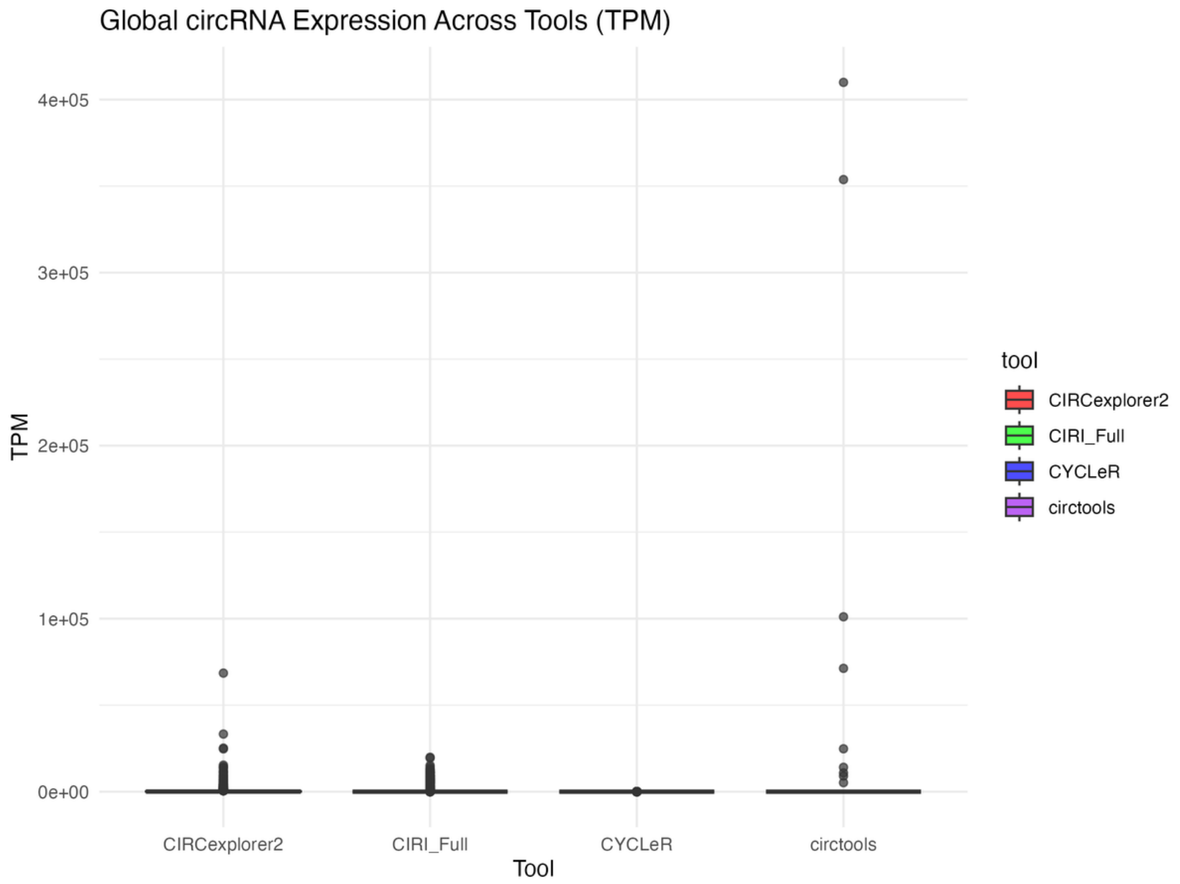


- CIRCexplorer2 detected the most circRNAs (1,373) – high sensitivity.
- CIRI-Full detected 97 circRNAs – conservative, focuses on full-length.
- CyCLeR detected 149 circRNAs – isoform-level accuracy
- circtools detected only 6 circRNAs – very strict criteria, high specificity.
- Only 1 circRNA was detected by all four tools – highest confidence candidate



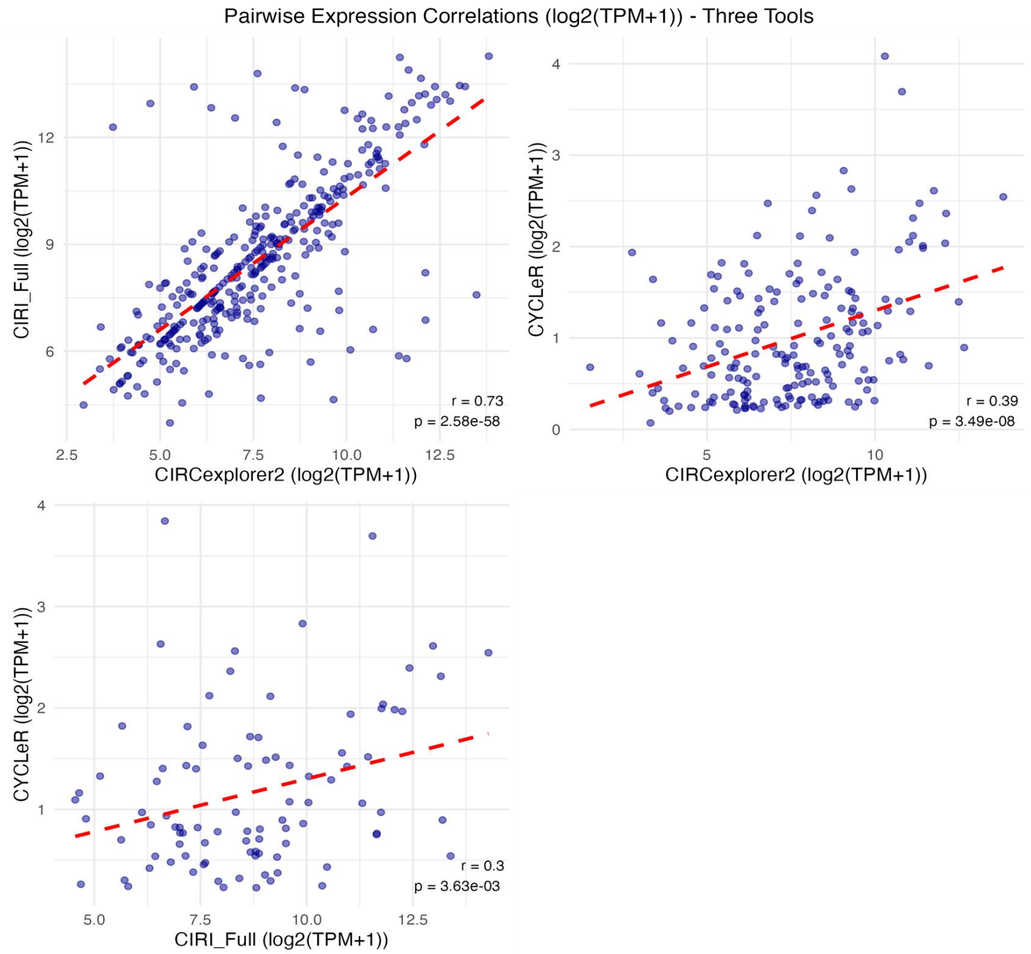
Results: circRNA Expression Differences

- CIRCexplorer2 – broadest expression range (low to very high).
- CIRI-Full – higher median expression, more concentrated.
- CyCLeR – lowest median expression (conservative quantification).
- circtools – bimodal pattern: one extremely high circRNA, others low.



Results: Correlation Analysis

- CIRCexplorer2 vs CIRI-Full – strong correlation (r = 0.73) – both alignment-based.
- CIRCexplorer2 vs CyCLeR – moderate correlation (r = 0.39) – different approaches.
- CIRI-Full vs CyCLeR – weak correlation (r = 0.30) – detect different subsets.



Strong correlation between alignment-based tools (CIRCexplorer2 & CIRI-Full). CyCLeR shows weaker correlation due to isoform-focused quantification.

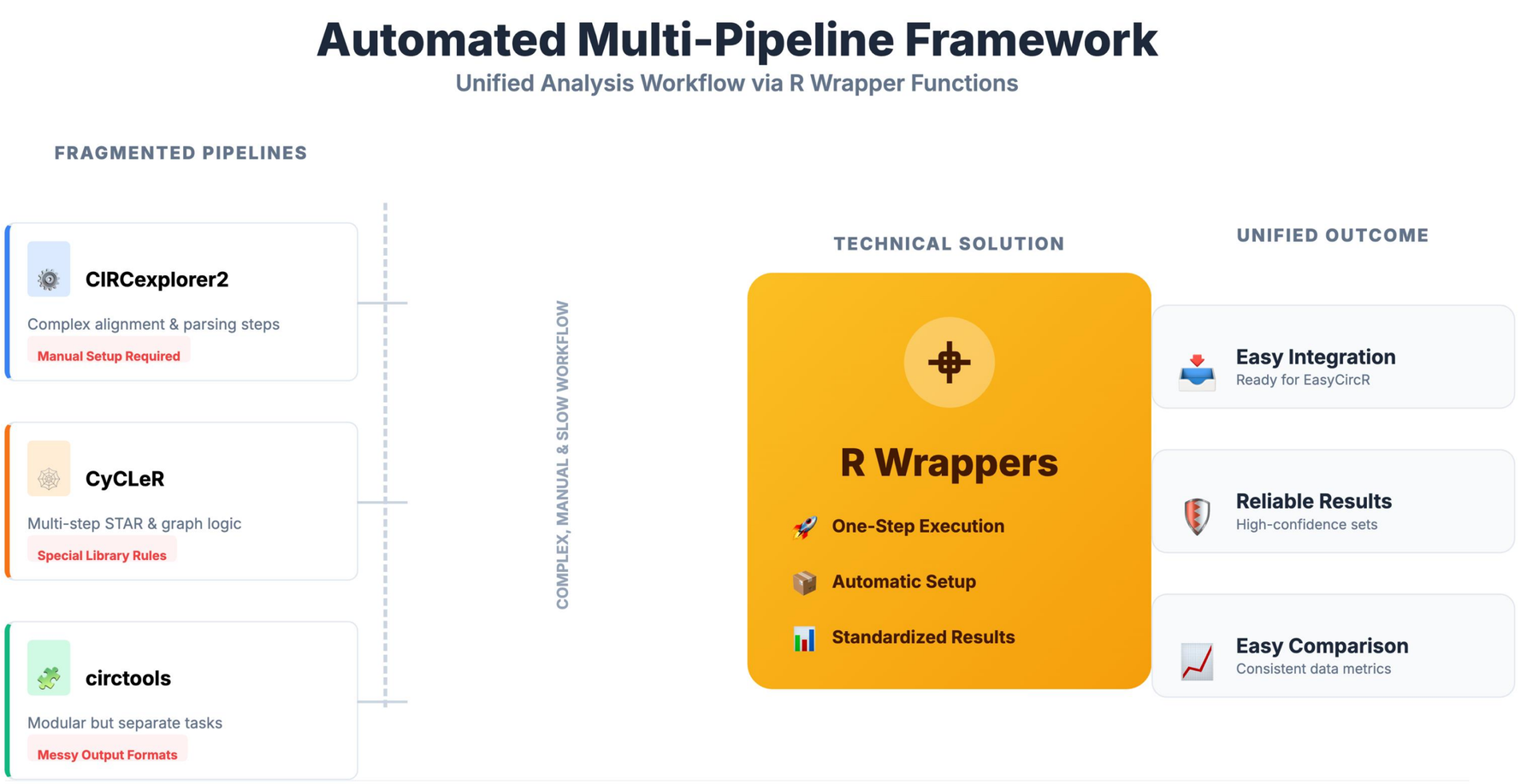
Results: Highest-Confidence circRNA – Detected by All Four Tools

- Expression estimates vary by up to 1,600-fold.
- circtools reports extremely high abundance; CyCLeR much lower.
- Despite quantitative differences, detection by all four tools confirms authenticity.
- Highlights how methodological choices affect abundance estimates.

circRNA Expression Comparison

Tool	Raw TPM	$\log_2$(TPM + 1)
CIRCexplorer2	88.9	6.49
CIRI-Full	208.0	7.71
CyCLeR	8.7	2.12
circtools	14,079.5	13.78

Automated Multi-Pipeline Framework



Discussion - Takeaways

Main insights from our comparison

1. No single tool is “best” – each has unique strengths:

- CIRCexplorer2 → detection
- CIRI-Full → full-length sequences
- CyCLeR → isoform accuracy
- circtools → validation and functional analysis

2. Tool choice drastically affects results – from 6 to 1,373 detections.
3. Expression estimates are not interchangeable – up to 1,600-fold differences.
4. Multi-tool consensus improves confidence – circRNAs detected by ≥3 tools are reliable candidates.

Conclusion & Future Work

1. Tool-Specific Biases
•Each circRNA detection pipeline shows unique biases.
•Affects both **circRNA identification** and **expression quantification**.
2. High-Confidence circRNAs
•Multi-pipeline consensus identifies a **robust set** of reliable circRNAs.
•Candidates for further **biological investigation**.
3. Strategic Pipeline Selection
•Choosing the right tool is critical.
•Must align with **experimental design** and **biological questions**.
4. Next Steps / Applications
•**Experimental Validation:** RT-qPCR, Northern blotting across cell lines/tissues.
•**Functional Characterization:** Study gene regulation, protein sponging, cellular processes.
•**Broader Applications:** Apply approach to other datasets, species, disease models, and clinical samples to discover **biomarkers and therapeutic targets**.

Thank you ☺